

7.2 Practice - Multiply and Divide

Simplify each expression.

$$1) \frac{8x^2}{9} \cdot \frac{9}{2}$$

$$3) \frac{9n}{2n} \cdot \frac{7}{5n}$$

$$5) \frac{5x^2}{4} \cdot \frac{6}{5}$$

$$7) \frac{7(m-6)}{m-6} \cdot \frac{5m(7m-5)}{7(7m-5)}$$

$$9) \frac{7r}{7r(r+10)} \div \frac{r-6}{(r-6)^2}$$

$$11) \frac{25n+25}{5} \cdot \frac{4}{30n+30}$$

$$13) \frac{x-10}{35x+21} \div \frac{7}{35x+21}$$

$$15) \frac{x^2-6x-7}{x+5} \cdot \frac{x+5}{x-7}$$

$$17) \frac{8k}{24k^2-40k} \div \frac{1}{15k-25}$$

$$19) (n-8) \cdot \frac{6}{10n-80}$$

$$21) \frac{4m+36}{m+9} \cdot \frac{m-5}{5m^2}$$

$$23) \frac{3x-6}{12x-24}(x+3)$$

$$25) \frac{b+2}{40b^2-24b}(5b-3)$$

$$27) \frac{n-7}{6n-12} \cdot \frac{12-6n}{n^2-13n+42}$$

$$29) \frac{27a+36}{9a+63} \div \frac{6a+8}{2}$$

$$31) \frac{x^2-12x+32}{x^2-6x-16} \cdot \frac{7x^2+14x}{7x^2+21x}$$

$$33) (10m^2+100m) \cdot \frac{18m^3-36m^2}{20m^2-40m}$$

$$35) \frac{7p^2+25p+12}{6p+48} \cdot \frac{3p-8}{21p^2-44p-32}$$

$$37) \frac{10b^2}{30b+20} \cdot \frac{30b+20}{2b^2+10b}$$

$$39) \frac{7r^2-53r-24}{7r+2} \div \frac{49r+21}{49r+14}$$

$$2) \frac{8x}{3x} \div \frac{4}{7}$$

$$4) \frac{9m}{5m^2} \cdot \frac{7}{2}$$

$$6) \frac{10p}{5} \div \frac{8}{10}$$

$$8) \frac{7}{10(n+3)} \div \frac{n-2}{(n+3)(n-2)}$$

$$10) \frac{6x(x+4)}{x-3} \cdot \frac{(x-3)(x-6)}{6x(x-6)}$$

$$12) \frac{9}{b^2-b-12} \div \frac{b-5}{b^2-b-12}$$

$$14) \frac{v-1}{4} \cdot \frac{4}{v^2-11v+10}$$

$$16) \frac{1}{a-6} \cdot \frac{8a+80}{8}$$

$$18) \frac{p-8}{p^2-12p+32} \div \frac{1}{p-10}$$

$$20) \frac{x^2-7x+10}{x-2} \cdot \frac{x+10}{x^2-x-20}$$

$$22) \frac{2r}{r+6} \div \frac{2r}{7r+42}$$

$$24) \frac{2n^2-12n-54}{n+7} \div (2n+6)$$

$$26) \frac{21v^2+16v-16}{3v+4} \div \frac{35v-20}{v-9}$$

$$28) \frac{x^2+11x+24}{6x^3+18x^2} \cdot \frac{6x^3+6x^2}{x^2+5x-24}$$

$$30) \frac{k-7}{k^2-k-12} \cdot \frac{7k^2-28k}{8k^2-56k}$$

$$32) \frac{9x^3+54x^2}{x^2+5x-14} \cdot \frac{x^2+5x-14}{10x^2}$$

$$34) \frac{n-7}{n^2-2n-35} \div \frac{9n+54}{10n+50}$$

$$36) \frac{7x^2-66x+80}{49x^2+7x-72} \div \frac{7x^2+39x-70}{49x^2+7x-72}$$

$$38) \frac{35n^2-12n-32}{49n^2-91n+40} \cdot \frac{7n^2+16n-15}{5n+4}$$

$$40) \frac{12x+24}{10x^2+34x+28} \cdot \frac{15x+21}{5}$$

$$41) \frac{x^2-1}{2x-4} \cdot \frac{x^2-4}{x^2-x-2} \div \frac{x^2+x-2}{3x-6}$$

$$43) \frac{x^2+3x+9}{x^2+x-12} \cdot \frac{x^2+2x-8}{x^3-27} \div \frac{x^2-4}{x^2-6x+9}$$

$$42) \frac{a^3+b^3}{a^2+3ab+2b^2} \cdot \frac{3a-6b}{3a^2-3ab+3b^2} \div \frac{a^2-4b^2}{a+2b}$$

$$44) \frac{x^2+3x-10}{x^2+6x+5} \cdot \frac{2x^2-x-3}{2x^2+x-6} \div \frac{8x+20}{6x+15}$$